

# Workshop 8 TEMPLATE

12 PM

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## Table of contents

|                                                                            |           |
|----------------------------------------------------------------------------|-----------|
| <b>1. Set up</b>                                                           | <b>2</b>  |
| <b>2. Clean data</b>                                                       | <b>3</b>  |
| <b>3. Visualizing data</b>                                                 | <b>5</b>  |
| What is the relationship between SLA and plant mass? . . . . .             | 5         |
| What are the differences in total mass between water treatments? . . . . . | 6         |
| What are the differences in total mass between species? . . . . .          | 7         |
| <b>4. Fitting models</b>                                                   | <b>9</b>  |
| Model fitting . . . . .                                                    | 10        |
| Model diagnostics . . . . .                                                | 11        |
| Model selection . . . . .                                                  | 12        |
| Stop and think: what does this model mean? . . . . .                       | 14        |
| <b>4. Model predictions</b>                                                | <b>15</b> |
| <b>5. Final figure</b>                                                     | <b>15</b> |
| <b>6. Writing</b>                                                          | <b>16</b> |
| Figure caption . . . . .                                                   | 16        |
| Basic components of test . . . . .                                         | 16        |
| Written communication . . . . .                                            | 17        |
| Methods . . . . .                                                          | 17        |
| Results . . . . .                                                          | 17        |

In this workshop, we will answer the question: How do specific leaf area, water treatment, and species influence plant mass?

**Specific leaf area** is a continuous variable measured in  $\text{cm}^2/\text{g}$ .

**Water treatment** is a categorical variable (i.e. a *factor*) with 2 levels: drought stressed (DS) and well watered (WW).

**Species** is a categorical variable (again, a *factor*) with 6 levels.

| Species name                      | Species code | Common name        |
|-----------------------------------|--------------|--------------------|
| <i>Encelia californica</i>        | ENCCAL       | Bush sunflower     |
| <i>Eschscholzia californica</i>   | ESCCAL       | California poppy   |
| <i>Penstemon centranthifolius</i> | PENCEN       | Scarlet bugler     |
| <i>Grindelia camporum</i>         | GRINCAM      | Gumweed            |
| <i>Salvia leucophylla</i>         | SALLEU       | Purple sage        |
| <i>Stipa pulchra</i>              | STIPUL       | Purple needlegrass |
| <i>Lotus scoparius</i>            | LOTSCO       | Deerweed           |

## 1. Set up

```
# read in your packages here
library(tidyverse) # general use
library(here) # file organization
library(janitor) # cleaning data frames
library(readxl) # reading excel files
library(scales) # modifying axis labels
library(ggeffects) # getting model predictions
library(MuMin) # model selection

# read in your data here

drought_exp <- read_xlsx(
  here("data", "Valliere_etal_EcoApps_Data.xlsx"),
  sheet = "First Harvest"

)
```

```
# storing colors to use for species
lotsco_col <- "#E69512"
pencen_col <- "#D6264F"
salleu_col <- "#6D397D"
```

```

enccal_col <- "#3A5565"
stipul_col <- "#3F564F"
esccal_col <- "#515481"
gricam_col <- "#6C91BD"

# storing colors to use for water treatments
ds_col <- "#A62F03"
ww_col <- "#045CB4"

# storing a ggplot theme (that will be used for all ggplots)
theme_set(theme_bw() +
  theme(text = element_text(size = 14)))

```

## 2. Clean data

Data source: Valliere, Justin; Zhang, Jacqueline; Sharifi, M.; Rundel, Philip (2019). Data from: Can we condition native plants to increase drought tolerance and improve restoration success? [Dataset]. Dryad. <https://doi.org/10.5061/dryad.v0861f7>

```

# creating new object
drought_exp_clean <- drought_exp |>
  # makes column names lowercase, and replaces spaces and underscores
  clean_names() |>
  # creative new column with full species names
  mutate(species_name = recode_values(
    species,
    "ENCCAL" ~ "Encelia californica", # bush sunflower
    "ESCCAL" ~ "Eschscholzia californica", # California poppy
    "PENCEN" ~ "Penstemon centranthifolius", # Scarlet bugler
    "GRICAM" ~ "Grindelia camporum", # Gumweed
    "SALLEU" ~ "Salvia leucophylla", # purple sage
    "STIPUL" ~ "Stipa pulchra", # purple needlegrass
    "LOTSCO" ~ "Lotus scoparius" # deerweed
  )) |>
  # creating new column of full names to water treatment types
  mutate(water_treatment = recode_values(
    water,
    "WW" ~ "Well watered",
    "DS" ~ "Drought stressed"
  )) |>
  # reordering the data

```

```
mutate(species_name = as_factor(species_name),
       species_name = fct_relevel(species_name,
                                   "Lotus scoparius",
                                   "Penstemon centranthifolius",
                                   "Salvia leucophylla",
                                   "Encelia californica",
                                   "Stipa pulchra",
                                   "Eschscholzia californica",
                                   "Grindelia camporum")) |>
mutate(water_treatment = as_factor(water_treatment),
       water_treatment = fct_relevel(water_treatment,
                                      "Drought stressed",
                                      "Well watered")) |>

# selecting columns of interest
select(species_name, water_treatment, sla, total_g)
```

Insert code to double check the structure of the data frame here.

```
str(drought_exp_clean)
```

```
tibble [70 x 4] (S3: tbl_df/tbl/data.frame)
 $ species_name   : Factor w/ 7 levels "Lotus scoparius",...: 4 4 4 4 4 4 4 4 4 4 ...
 $ water_treatment: Factor w/ 2 levels "Drought stressed",...: 2 2 2 2 2 1 1 1 1 1 ...
 $ sla            : num [1:70] 170 215 209 216 222 ...
 $ total_g        : num [1:70] 0.455 0.329 0.45 0.359 0.352 ...
```

Insert code to display 10 random rows from the data frame here.

```
slice_sample(
  drought_exp_clean, # clean data frame to display
  n=10 #show 10 random rows
)
```

```
# A tibble: 10 x 4
  species_name      water_treatment    sla total_g
  <fct>            <fct>          <dbl>  <dbl>
1 Salvia leucophylla Drought stressed  96.1  0.278
2 Lotus scoparius    Drought stressed  91.4  0.0626
3 Salvia leucophylla Well watered      113.  0.195
4 Grindelia camporum Drought stressed 172.  0.333
```

|    |                          |                  |      |       |
|----|--------------------------|------------------|------|-------|
| 5  | Eschscholzia californica | Drought stressed | 130. | 0.157 |
| 6  | Eschscholzia californica | Drought stressed | 105. | 0.231 |
| 7  | Eschscholzia californica | Well watered     | 116. | 0.423 |
| 8  | Grindelia camporum       | Well watered     | 205. | 0.412 |
| 9  | Grindelia camporum       | Drought stressed | 164. | 0.428 |
| 10 | Encelia californica      | Well watered     | 209. | 0.450 |

### 3. Visualizing data

#### What is the relationship between SLA and plant mass?

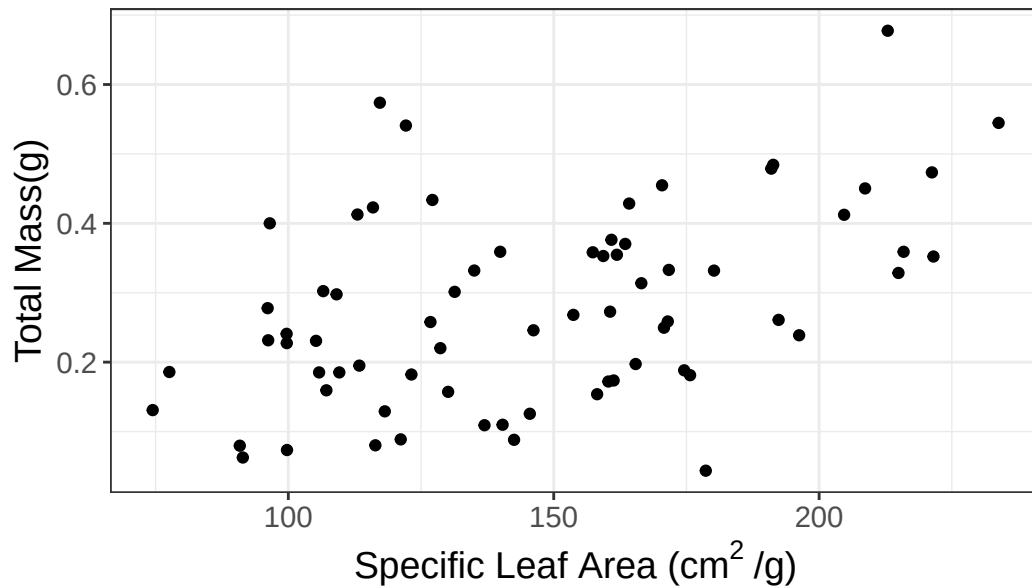
Make a scatterplot to visualize SLA (the predictor) on the x-axis, and total mass (the response) on the y-axis.

Label the axes with units where appropriate.

Add a title describing the relationship (e.g. as SLA increases, total mass \_\_\_\_\_).

```
#base layer: ggplot
ggplot(data = drought_exp_clean,
       mapping = aes(
         #x-axis: specific leaf area
         #y-axis: total plant mass
         x = sla,
         y = total_g)) +
#scatter plot
geom_point() +
labs(x = expression("Specific Leaf Area (cm"2~/g)"),
     y = "Total Mass(g)",
     title = "As SLA Increases, Total Mass Increases"
    )
```

## As SLA Increases, Total Mass Increases



### What are the differences in total mass between water treatments?

Make a jitterplot to visualize the differences in plant mass between water treatments.

Label the axes with units where appropriate.

Add a title describing the differences in (mean or median) plant mass between water treatments.

```
#base layer: ggplot
ggplot(data = drought_exp_clean,
       #x-axis: water treatment
       #y-axis: total plant mass
       mapping = aes(x = water_treatment,
                     y = total_g,
                     color = water_treatment)) +

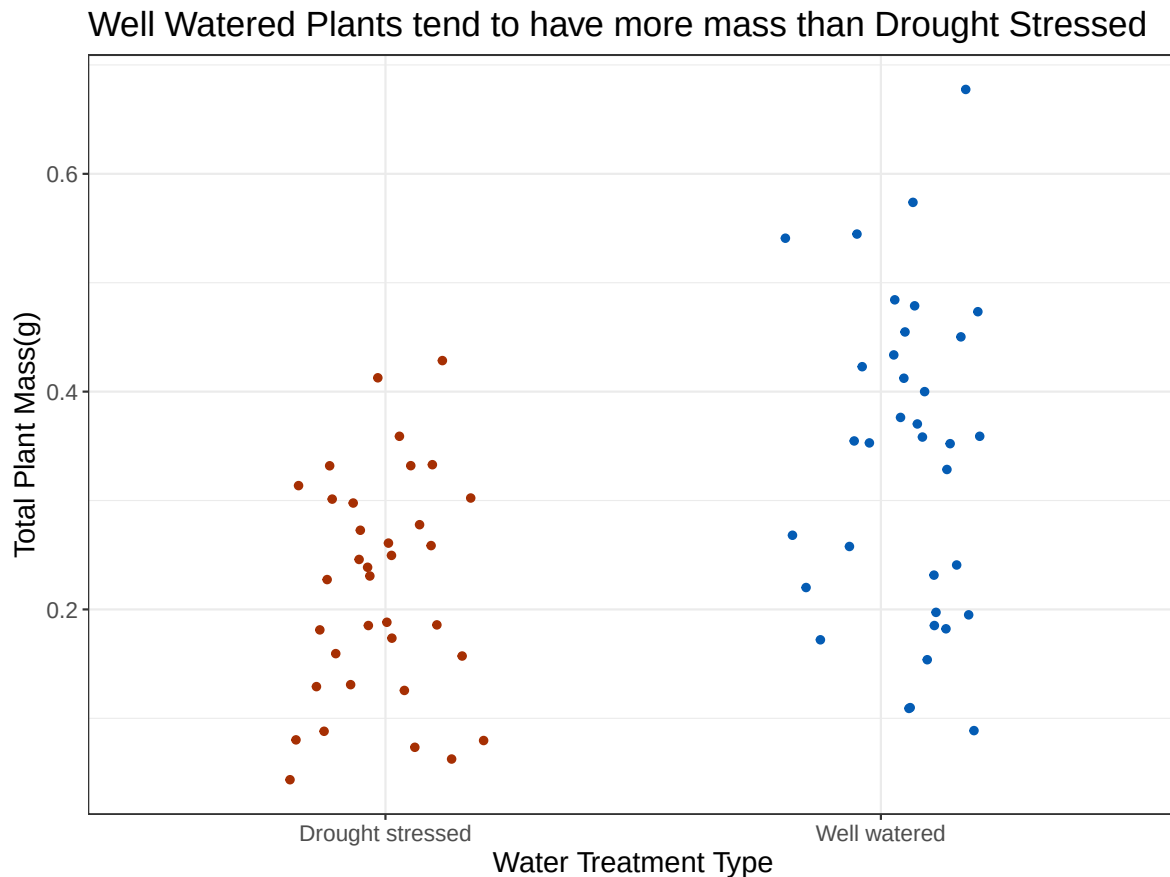
#jitter plot
geom_jitter(height = 0,
            width = 0.2) +

#changing colors
scale_color_manual(values = c(
  "Drought stressed" = ds_col,
  "Well watered" = ww_col
```

```

)) +
#relabeling
labs(x = "Water Treatment Type",
     y = "Total Plant Mass(g)",
     title = "Well Watered Plants tend to have more mass than Drought Stressed") +
#removing legend
theme(legend.position = "none")

```



### What are the differences in total mass between species?

Make a jitterplot to visualize the differences in plant mass between species.

Label the axes with units where appropriate.

Add a title describing the differences in (mean or median) plant mass between species.

Bonus: order the x-axis by decreasing mean plant mass.

```

#base layer: ggplot
ggplot(data = drought_exp_clean,
      #x-axis: species
      #y-axis: plant mass
      #coloring by species
      mapping = aes(x = species_name,
                    y = total_g,
                    color = species_name)) +

#jitterplot
geom_jitter(height = 0,
            width = 0.2) +

#wrapping x-axis text
scale_x_discrete(labels = label_wrap(10)) +

#relabelling axes
labs(x = "Species Name",
     y = "Total Plant Mass(grams)",
     title = "Species differ in plant mass") +

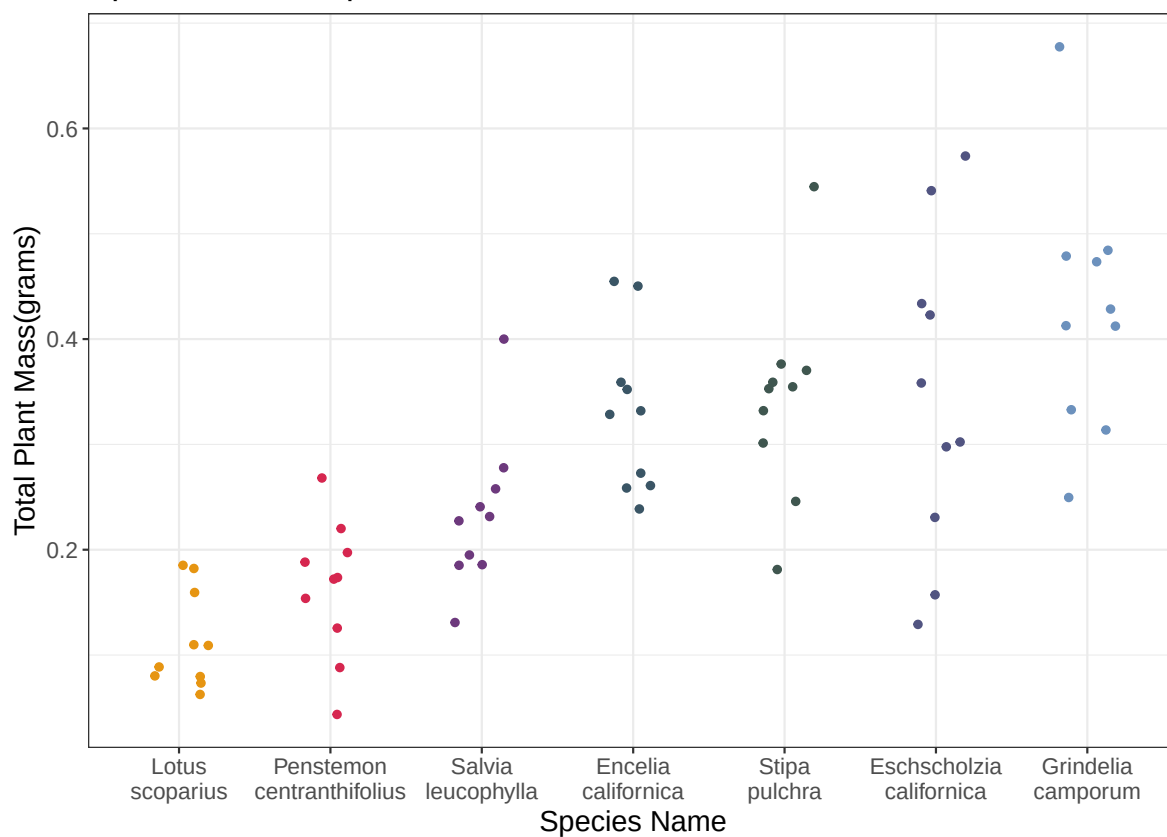
#using custom colors
scale_color_manual(
  values = c(
    "Lotus scoparius" = lotsco_col,
    "Penstemon centranthifolius" = pencen_col,
    "Salvia leucophylla" = sallev_col,
    "Encelia californica" = enccal_col,
    "Stipa pulchra" = stipul_col,
    "Eschscholzia californica" = esccal_col,
    "Grindelia camporum" = gricam_col
  )
) +

#removing legend
theme(legend.position = "none")

```



## Species differ in plant mass



## 4. Fitting models

8 models total:

| Model number | SLA | Water treatment | Species | Predictor list              |
|--------------|-----|-----------------|---------|-----------------------------|
| 0            |     |                 |         | no predictors (null model)  |
| 1            | X   | X               | X       | all predictors (full model) |
| 2            | X   | X               |         | SLA and water treatment     |
| 3            | X   |                 | X       | SLA and species             |
| 4            |     | X               | X       | water treatment and species |
| 5            | X   |                 |         | SLA                         |
| 6            |     | X               |         | water treatment             |
| 7            |     |                 | X       | species                     |

## Model fitting

```
# model 0: null model
model0 <- lm(
  total_g ~ 1,
  data = drought_exp_clean
)

# model 1: all predictors
model1 <- lm(
  total_g ~ sla + water_treatment + species_name,
  data = drought_exp_clean
)

# model 2: SLA and water treatment
model2 <- lm(
  total_g ~ sla + water_treatment,
  data = drought_exp_clean
)

# model 3: SLA and species
model3 <- lm(
  total_g ~ sla + species_name,
  data = drought_exp_clean
)

# model 4: water treatment and species
model4 <- lm(
  total_g ~ water_treatment + species_name,
  data = drought_exp_clean
)

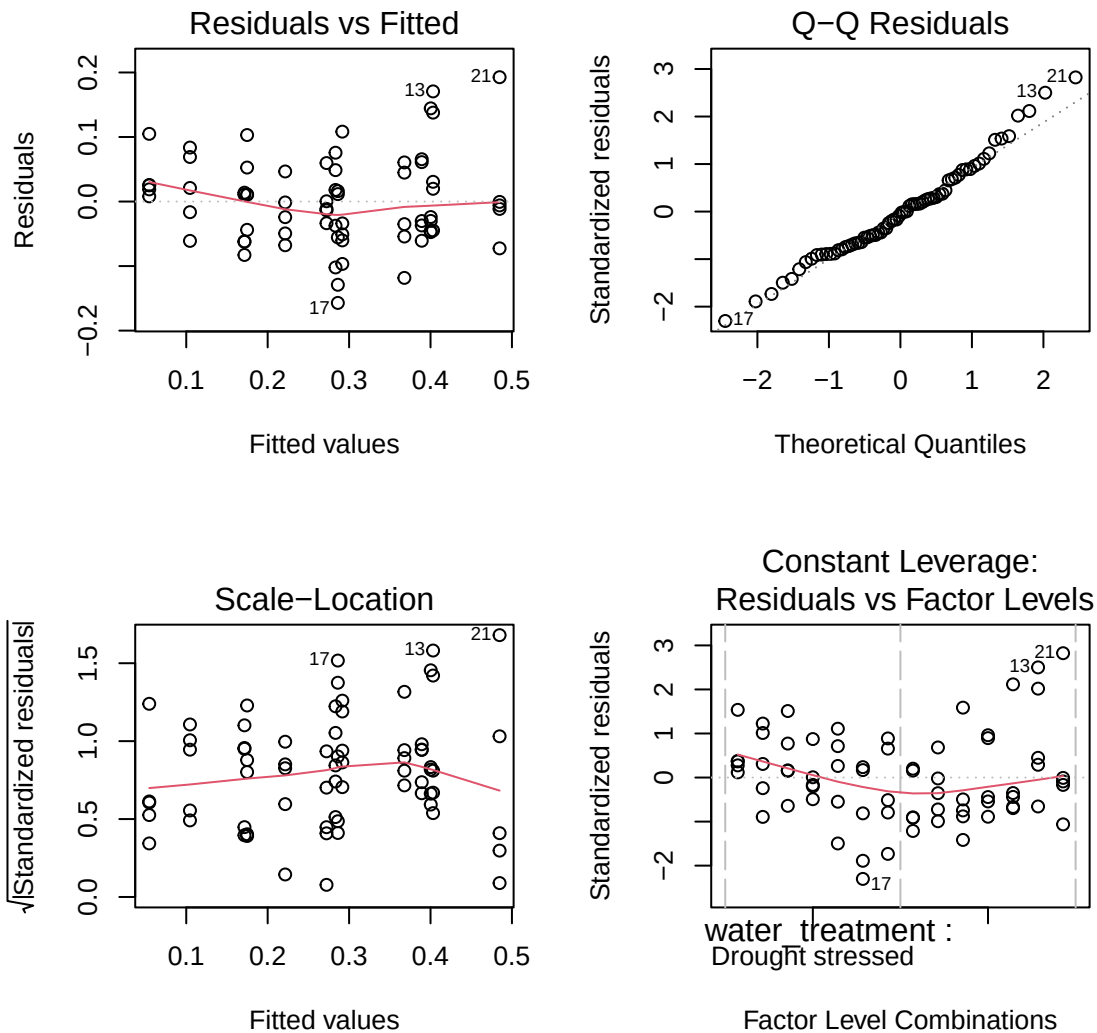
# model 5: SLA
model5 <- lm(
  total_g ~ sla,
  data = drought_exp_clean
)

# model 6: water treatment
model6 <- lm(
  total_g ~ water_treatment,
  data = drought_exp_clean
)
```

```
)  
  
# model 7: species  
model7 <- lm(  
  total_g ~ species_name,  
  data = drought_exp_clean  
)
```

## Model diagnostics

```
#displaying plots in 2x2 grid  
par(mfrow = c(2,2))  
#plotting diagnostics  
plot(model4)
```



- residuals are normally distributed (QQ Plot)
- residuals are homoscedastic (Residuals vs Fitted/Scale-Location)
- no outliers that could be influencing model estimates (from our leverage plot)

## Model selection

Insert code to use the `AICc()` function to calculate the AIC for each model.

Arrange by decreasing AIC.

```
AICc(
  model0,
  model1,
  model2,
  model3,
  model4,
  model5,
  model6,
  model7

) |>
  arrange(AICc)
```

|         | df | AICc       |
|---------|----|------------|
| model4  | 9  | -156.19595 |
| model11 | 10 | -153.75361 |
| model13 | 9  | -124.07569 |
| model17 | 8  | -120.30191 |
| model12 | 4  | -95.82521  |
| model15 | 3  | -88.85180  |
| model16 | 3  | -86.76661  |
| model10 | 2  | -74.98036  |

Insert code to display the summary for the best model.

```
summary(model4)
```

Call:

```
lm(formula = total_g ~ water_treatment + species_name, data = drought_exp_clean)
```

Residuals:

| Min       | 1Q        | Median    | 3Q       | Max      |
|-----------|-----------|-----------|----------|----------|
| -0.157087 | -0.046953 | -0.003733 | 0.041244 | 0.192657 |

Coefficients:

|                             | Estimate | Std. Error | t value | Pr(> t )     |
|-----------------------------|----------|------------|---------|--------------|
| (Intercept)                 | 0.05455  | 0.02451    | 2.225   | 0.02973 *    |
| water_treatmentWell watered | 0.11695  | 0.01733    | 6.746   | 5.90e-09 *** |

```

species_namePenstemon centranthifolius 0.05003 0.03243 1.543 0.12799
species_nameSalvia leucophylla 0.12020 0.03243 3.706 0.00045 ***
species_nameEncelia californica 0.21774 0.03243 6.714 6.70e-
09 ***
species_nameStipa pulchra 0.22881 0.03243 7.055 1.72e-
09 ***
species_nameEschscholzia californica 0.23164 0.03243 7.143 1.22e-
09 ***
species_nameGrindelia camporum 0.31335 0.03243 9.662 5.53e-
14 ***
---
```

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.07252 on 62 degrees of freedom

Multiple R-squared: 0.7535, Adjusted R-squared: 0.7257

F-statistic: 27.08 on 7 and 62 DF, p-value: < 2.2e-16

## Stop and think: what does this model mean?

### What is the best model?

The best model includes water treatment and species as predictors

### How much variation in the response (total mass, in grams) does this model explain?

72.6% of the variation in plant mass is explained by this model.

### How do we interpret the effects of the predictors on the response variable (again, total mass in grams)?

-Species influences plant mass Treatment influences plant mass

- (Intercept): The intercept is the model prediction for plant mass for Lotus scoparius in the Drought stressed treatment.
- water\_treatmentWell watered: model prediction for the difference in Lotus scoparius mass between well watered and drought stressed treatments.
- species\_namePenstemon centranthifolius: difference between Penstemon mass in the well watered treatment and Lotus scoparius mass in the well watered treatment
- species\_nameEschscholzia californica: difference between Eschscholzia californica mass in the well watered treatment vs Lotus scoparius mass in the well watered treatment

## 4. Model predictions

Insert code to generate model predictions for the best model.

Rename the `x` column and `group` column to match the original data frame (`drought_exp_clean`).

Look at the object before plotting.

```
preds <- ggpredict(  
  model4, #model object  
  terms = c("water_treatment", "species_name") #predictors  
) |>  
  #renaming columns to original data frame  
  rename(water_treatment = x,  
         species_name = group)
```

## 5. Final figure

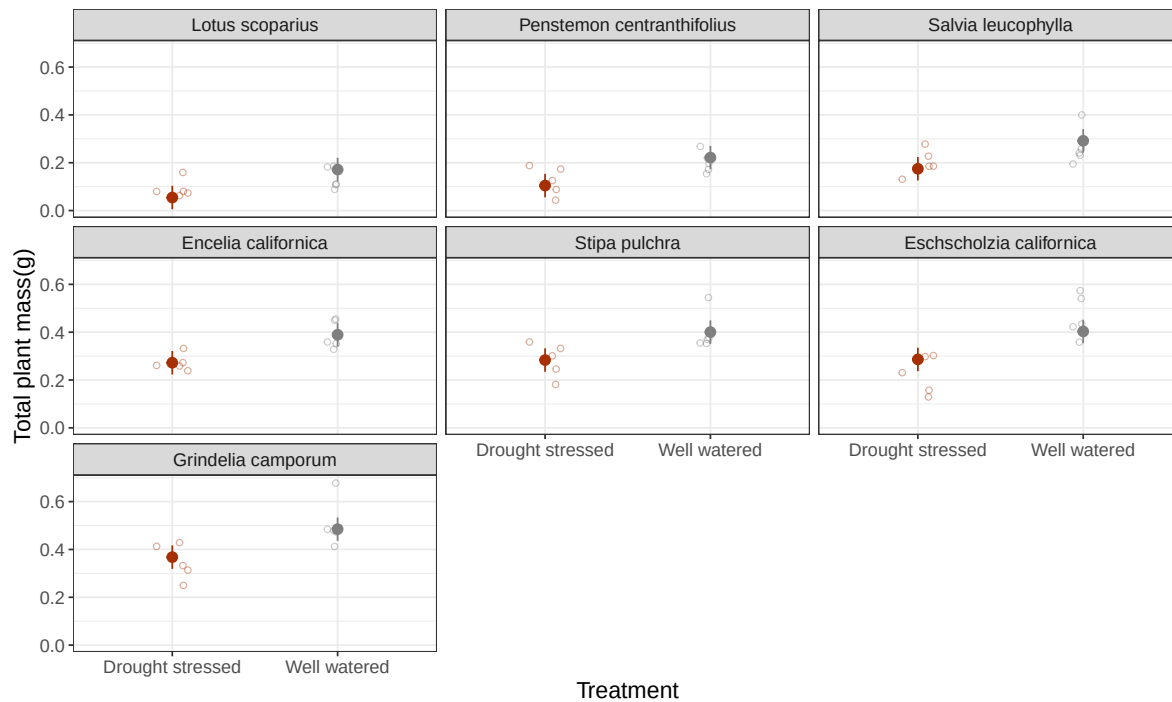
Create a clean, publication ready figure depicting predictions for the best model and the underlying data.

```
#base layer: ggplot  
ggplot(data = drought_exp_clean,  
       #x-axis: treatment  
       #y-axis: plant mass  
       #color by water treatment  
       mapping = aes(x = water_treatment,  
                     y = total_g,  
                     color = water_treatment)) +  
  #jitter: represents individual plants for each treatment in each species  
  geom_jitter(height = 0,  
             width = 0.1,  
             alpha = 0.4,  
             shape = 21) +  
  #model predictions with 95% CI  
  geom_pointrange(data = preds,  
                 mapping = aes(x = water_treatment,  
                               y = predicted,  
                               ymin = conf.low,  
                               ymax = conf.high)) +  
  #creates different panels for each species
```

```

facet_wrap(~ species_name) +
  #adds custom colors
  scale_color_manual(values = c(
    "Drought stressed" = ds_col,
    "Well Watered" = ww_col
  )) +
  #removes legends
  theme(legend.position = "none") +
  #relabel axes
  labs(x = "Treatment",
       y = "Total plant mass(g)")

```



## 6. Writing

### Figure caption

### Basic components of test

### predictors



response

test

distribution

parameters (degrees of freedom)

$R^2$

p-value

significance level

reference level(s)

## **Written communication**

### **Methods**

We used linear models to determine the predictive relationship between specific leaf area ( $\text{cm}^2/\text{g}$ ), treatment, and species on plant mass (g). We evaluated model assumptions of normally distributed errors using QQ plots of model residuals. We evaluated assumptions of constant variance of errors using Residuals vs Fitted and Scale-Location plots. We used Akaike's Information Criterion (AIC) to select the simplest model that explained the most variation in plant mass.

### **Results**

We found that the simplest model describing the greatest variation in plant mass included treatment and species as predictors (linear model:  $F(2, 67) = 14.57$ ,  $p < 0.001$ ,  $\alpha = 0.05$ ).